METRIC

MIL-S-12060D
29 February 1984
SUPERSEDING
MIL-S-12060C
13 January 1977

MILITARY SPECIFICATION

SULFUR MONOCHLORIDE, TECHNICAL (METRIC)

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one technical grade of sulfur monochloride (S_2Cl_2).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Armament Research and Development Center, US Army Armament, Munitions and Chemical Command, ATTN: DRSMC-TSC-S(A), Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6810

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SPECIFICATIONS

FEDERAL

NN-P-71	-	Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
PPP-B-585	-	Boxes, Wood, Wirebound
PPP-C-186	-	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals
PPP-C-2020	-	Chemicals, Liquid, Dry, and Paste: Packaging of
PPP-F-320	-	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-1168	-	Ammunition Lot Numbering

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171 to 179	-	Department of Transportation Hazardous Materials Regulations

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders for the above publication should cite "49 CFR 171 to 179.")

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

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ASTM STANDARDS

D 1193 - Reagent Water
D 2111 - Specific Gravity of Hallogenated Organic Solvents and Their Admixtures
E 256 - Chlorine in Organic Compounds by Sodium Peroxide Bomb Ignition
E 257 - Sulfur in Organic Compounds by Sodium Peroxide Bomb Ignition

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Sulfur content. Sulfur monochloride shall contain no less than 47.0 percent by weight sulfur when tested as specified in 4.2.4.1.

3.2 Chlorine content. Sulfur monochloride shall contain no less than 52.0 ercent by weight chlorine when tested as specified in 4.2.4.2.

3.3 Specific gravity. Sulfur monochloride shall have a specific gravity at 20° ± 1°C of 1.680 to 1.690 when tested as specified in 4.2.4.3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the sulfur monochloride produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3). Each lot shall be identified and controlled in accordance with MIL-STD-1168.

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4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For sulfur monochloride test. See 6.5 for sampling and testing precautions. Sampling shall be conducted in accordance with table I. A representative specimen of approximately 200 grams (g) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

TABLE I. Sampling for sulfur monochloride test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.2.3 For container leakage test. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled unit, intermediate, or shipping container, as applicable, ready for shipment. Sample unit, intermediate and shipping containers shall be examined for the following defects using an AQL of 1.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Fiberboard liner, pads, or partitions missing or not as specified (when required)
(f) Unitization not as specified
(g) Marking incorrect, missing, or illegible

4.2.3.2 For sulfur monochloride test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

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.2.3.3 For container leakage test. The sample unit shall be one container. The sample containers selected in 4.2.2.3 shall be tested as specified in 4.2.5 using an AQL of 1.0 percent defective.

4.2.4 Sulfur monochloride tests. See 6.5 for sampling and testing precautions. Water in accordance with ASTM D 1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Sulfur content. Determine the percent by weight sulfur in the specimen in accordance with ASTM E 257.

4.2.4.2 Chlorine content. Determine the percent by weight chlorine in the specimen in accordance with ASTM E 256.

4.2.4.3 Specific gravity. Determine the specific gravity of the specimen in accordance with method B of ASTM D 2111 using an ASTM No. 138H hydrometer.

4.2.5 Container leakage test. Place the container in each of the following positions, and leave it in each for a period of 15 minutes:

(a) Upright
(b) Upside down
(c) On one side (or one quadrant)
(d) One one end (or second quadrant)
(e) On other side (or fourth quadrant)

Examine the container after each period for any evidence of leakage.

5. PACKAGING

5.1 Preservation. Sulfur monochloride shall be preserved level A, B or C as specified (see 6.2), and shall be in accordance with the general requirements of PPP-C-2020 and Department of Transportation (DOT) regulations.

5.1.1 Level A.

5.1.1.1 Unit packing. A quantity of 500 (+5 or -0) g of sulfur monochloride shall be unit packed level A in a bottle conforming to group A, class 1, type of glass optional, style 1 narrow-mouth, grade 1, light-resistant with screw-cap A or R and outer seal A of PPP-C-186. The screw-cap shall be closed on the bottle to a torque within the range specified by the bottle manufacturer. There shall be no evidence of leakage when tested as specified in 4.2.5. Each bottle shall be overpacked with closure end uppermost as specified for enclosure B in PPP-C-2020.

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5.1.1.2 Intermediate packing. Twelve unit packs (bottles) of sulfur monochloride shall be intermediately packed with closures uppermost in three rows of four bottles per row in a close-fitting wirebound wood box. The box shall conform to class 3, style optional of PPP-B-585. Each bottle shall be placed in a close- fitting cell formed by half-slotted partitions formed from fiberboard. All inside faces of the box shall be lined with fiberboard. Fiberboard pads shall be inserted where needed to prevent motion of contents. Fiberboard shall conform to grade W5c or W6c of PPP-F-320. The box shall be closed as specified for the closure DOT regulations, title 49 paragraph 173.185.

5.1.2 Level B.

5.1.2.1 Unit packing. Sulfur monochloride shall be unit packed level B in the same manner as is specified for level A.

5.1.2.2 Intermediate packing. Sulfur monochloride shall be intermediately packed in the same manner as is specified for level A, except that the liner, partitions and pads shall conform to class domestic, variety SW or DW, minimum grade 175 of PPP-F-320.

5.1.3 Level C. A quantity of 500 (+5 or -0) g of sulfur monochloride shall be unit packed level C in a manner to assure specified purity and quantity of the sulfur monochloride from supply source to the initial intended destination for a minimum duration of six months. The container shall be a narrow-mouth glass bottle with screw-cap highly resistant to sulfur monochloride. The bottle shall be placed and centered with closure-end uppermost in a tin-plate can and cushioned therein with resilient noncombustible absorbent material. The can shall be sealed closed. There shall be no leakage from bottle or can when tested as specified in 4.2.5.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2), and shall be in accordance with the general requirements of PPP-C-2020 and DOT regulations.

5.2.1 Level A. Two intermediate packs of sulfur monochloride shall be packed level A, with bottle closures uppermost, in a close-fitting wirebound wood box conforming to class 3, style optional with veneer faceboards of PPP-B-585. All inside faces of the box shall be lined with fiberboard. Motion of contents shall be prevented by inserting fiberboard pads as needed. Liner and pads shall be formed of material conforming to grade V3c of PPP-F-320. The box shall be closed as specified in PPP-B-585.

5.2.2 Level B. Two intermediate packs of sulfur monochloride shall be packed level B, with bottle closures uppermost, in a close-fitting wirebound box conforming to class 2, style optional, with veneer faceboards of PPP-B-585. All inside faces of the box shall be lined with fiberboard. Motion of contents shall be prevented by inserting fiberboard pads as needed. The liner and pads shall be formed from material conforming to grade V3c of PPP-F-320. The box shall be closed as specified in PPP-B-585.

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5.2.3 Level C. Two intermediate packs of sulfur monochloride shall be packed in a close-fitting wood box conforming to DOT regulations. The sulfur monochloride shall be packed in a manner to assure acceptance by common carrier and conformance with Uniform Freight Classification Rules and National Motor Freight Classification Rules.

5.3 Unitization. Uniform quantities of level A, B or C packs shall be palletized in accordance with load type I of MIL-STD-147, using the softwood pallet conforming to type IV of NN-P-71.

5.4 Marking. Level A, B and C containers and pallet loads shall be marked in accordance with MIL-STD-129 and DOT regulations. Each container shall also be marked to show the date of manufacture and lot or batch number of the sulfur monochloride, and the following precautionary information:

WARNING!
CORROSIVE! Causes burns.

Very irritating to eyes, skin and respiratory tract. Avoid breathing vapors. Avoid contact with eyes, skin or clothing. If contact occurs, flush with copious amounts of water (in eyes, flush with water for 15 minutes holding eyelids open). If taken internally, drink large amounts of water or milk. Do not get water in container. Seek medical attention immediately.

Provide and require the use of personal protective equipment as designated by local industrial hygiene, medical and safety personnel.

6. NOTES

6.1 Intended use. Sulfur monochloride is intended for use in the manufacture of chemical agents.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification,
(b) Level of preservation and packing required (see 5.1 and 5.2).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E 29.

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6.5 Sampling and testing precautions. Undue exposure of sulfur monochloride to the atmosphere should be avoided. The material fumes in moist air and is extremely irritating. Necessary precautions should be observed in sampling and handling the material. Consult industrial hygiene personnel for guidance.

Custodian:

Army - EA

Review activities:

Army - MD
DLA - GS

Preparing activity:

Army - EA

Project No. 6810-B422

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-12060D	SULFUR MONOCHLORIDE, TECHNICAL (METRIC)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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